



OPEN Boron toxicity affects plant growth by destroying the cell membrane stability and micronutrient balance in melon

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This study focused on the toxic effects of boron (B) on the growth and micronutrient composition of melon plants. The seedlings were grown in pots with different concentrations of B (0, 20, 40, 60, and 80 mg B/kg soil) for 30 days. Plant height, seedling fresh and dry weight, leaf number, area, temperature, dry matter, chlorophyll index (Chl), relative water content (RWC), and cell membrane stability (CMS) were measured. B, Fe, Mn, Zn, and Cu in the shoots and soil B were determined. Increasing B linearly decreased plant height, while 20 mg B/kg stimulated fresh weight, leaf number, and area. However, seedling growth was inhibited at concentrations higher than 40 mg B/kg. The RWC and Chl were not changed between the control and 60 mg B/kg. Conversely, applying 20 or 40 mg B/kg improved the CMS. Increasing B caused a linear increase in B in plant tissues while disturbing the ionic balance with Fe, Mn, Zn, and Cu. It was concluded that increasing B caused a detrimental reduction in Chl, RWC, CMS, and Fe concentration, whereas the leaf temperature increased, resulting in the inhibition of root, stem, and leaf growth in melon plants.

Keywords *Cucumis melo* L., Boron, Toxicity, Ion balance, Electrolyte leakage

Melon, scientifically known as *Cucumis melo* L., is a warm-season vegetable with juicy and sweet fruits grown in Türkiye. There is a wide variety of melon cultivars with different fruit types, sizes, and colors¹. It is commonly grown as a cantaloupe or muskmelon with a creeping stem. Edible mature or fresh fruits are also traditionally consumed as a vegetable and for making pickles^{2,3}. Based on 2023 data, its production area is 63,000 hectares in 71 out of 81 provinces in Türkiye, resulting in a production of 1,587,230 tons⁴. Most of the melon production has taken place in open fields without irrigation, relying solely on rainfall, in arid and semiarid areas with favorable climatic conditions for melon cultivation.

Boron (B) is an important micronutrient that plays a critical role in physiological processes in plants such as cell elongation, sugar transport, meristematic tissue development, and cell wall synthesis⁵. The range between deficiency and toxicity levels of B is very narrow^{6,7} because the critical levels depend on several factors such as soil type, pH, water status, texture, air humidity, temperature, plant species, and genotype⁸. The concentration of B in soils varies greatly with irrigation regime, and boron deficiency exists in humid regions due to high rainfall, resulting in B-leaching from the root zone^{9–11}. On the other hand, B toxicity occurs in different regions of the world and is a common problem in arid and semi-arid regions where the annual rainfall is insufficient to dissolve B in the soil^{5,12}. B-containing fertilizers and irrigation water can also cause boron toxicity^{13,14}. B concentrations in soils range from 10 to 300 mg/kg of soil, but B concentrations in surface water vary between 0.001 and 2 mg/L¹⁵. A study reported that soil B concentrations greater than 12 mg/kg were toxic to several plants in regions with less than 550 mm of annual rainfall¹⁶. In Türkiye, 46.2% of the soils contain excessive boron and 3.3% suffer from boron toxicity¹⁷.

Plant species exhibit varying degrees of tolerance to B toxicity¹⁸, depending on their ability to maintain low B concentrations in shoots. B-tolerant varieties are able to exclude B from their roots more efficiently and transport less B to their shoots. Plants exposed to excessive boron exhibit distinct visual symptoms, such as stunted growth, yellowing, and drying of tips and margins of older leaves¹⁹. Additionally, the plants growing in soil with high B concentrations presented high B contents in their seeds, negatively affecting germination ability²⁰. Melon is classified as a moderately B-tolerant plant²¹. In a recent study, it was reported that higher levels of 20 mg B/L adversely affected seedling growth but did not have toxic effects on melon germination²². However,

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melon was more sensitive to boron toxicity than squash, but more tolerant than cucumber and corn²³. A study showed that the grafted plants had a higher fruit yield than the non-grafted plants when boron accumulation was present in the leaves¹⁴. Conversely, the beneficial effects of boron application on melon seedlings under boron-deficient conditions were demonstrated by²⁴, who observed an increase in the leaf chlorophyll content and soluble solids in the fruit. The present study aimed to determine the toxic effects of B on the growth of melon seedlings by evaluating their morphological and physiological parameters, as well as micronutrient contents, under laboratory conditions.

Materials and methods

A laboratory experiment was conducted to determine the toxicity level of boron (B) on the seedlings of melon by assessing agro-morphologic, physiologic, and mineral contents during the early growth stage. Field soil was collected from the experimental fields of Eskişehir Osmangazi University at a depth of 0–20 cm. The structure of the field soil was identified as clay-loam with low organic matter (1.1%), high potassium content, slightly alkaline (pH=7.8), and ECe=0.8 dS m⁻¹ (Electrical conductivity of saturation extract of the soil). Melon seedlings at the V₂ stage from cultivar Pirlanta, purchased from a commercial seedling producer in Eskişehir, Türkiye, were used in the experiment. The seedlings were transplanted directly into 0.5 L plastic pots containing a 3:1 mixture of field soil and sand, following the application of overdosing B levels of 20, 40, 60, and 80 mg B per kg of soil. Sodium borate (Etidot-67, 20.9% Na₂B₄O₇·4H₂O) served as the boron source, mixed homogeneously to induce toxicity symptoms during a short period of exposure. Basal fertilization was applied twice with 50 mL of liquid N-P-K (8-8-8) with a concentration of 0.2% to each pot after transplanting and one week later, up to 80% of the field capacity. The seedlings were grown in a growth chamber at 23/18 °C day/night for a 16/8 h photoperiod with a relative humidity of 65–75%. The light intensity was 20,000 lx during the experiment. The pots were irrigated with tap water daily to maintain the soil water content at field capacity.

Morphological characteristics

Thirty days after the boron treatments, the plants were harvested, and the following measurements were performed. Plant height was measured as the distance from the soil surface to the top of the melon seedlings. The number of leaves on each plant was counted. The shoot parts of the plants were cut above the soil surface and weighed directly to determine seedling fresh weight (SFW). They were then dried at 80 °C for 24 h to determine seedling dry weight (SDW). Dry matter was calculated by dividing SDW by SFW as a percentage. Leaf area was measured on scanned leaves using the Image J program²⁵.

Physiological characteristics

Chlorophyll index and leaf temperature were measured using a chlorophyll meter SPAD-502 (Konica Minolta Corporation, Osaka, Japan) as the SPAD index, and an infrared thermometer (Trotec BP21, Germany), respectively.

Relative water content (RWC) of leaves was calculated using the formula described by²⁶: $RWC (\%) = [(FW - DW) / (TW - DW)] \times 100$, where FW = fresh weight, DW = dry weight, and TW = turgid weight of the leaf.

Cell membrane stability (CMS) was evaluated by measuring electrolyte leakage. It was determined on the second leaf from the top of the plants. After washing the leaves with deionized water to remove contaminations on the leaf surface, six discs of 10 mm diameter were excised from the leaves, weighed, and placed in a glass tube filled with 20 mL of deionized water. At the end of the 24-hour incubation period at 20 °C, the initial EC (Li) of the solution was read directly with an EC meter (WTW 3.15i, Germany). They were then incubated in a water bath at 90 °C for 60 min, and the EC (Lo) was read again after cooling it to room temperature. It was calculated as $(Li / Lo) \times 100$ and expressed as a percentage. Higher electrolyte leakage indicates lower cell membrane stability.

Mineral analysis

The concentrations of B, Mn, Fe, Zn, and Cu in the shoot of melon seedlings and the soil B concentration were determined by ICP-OES (Inductively Coupled Plasma Optical Emission Spectrometer, Agilent's 5110). After washing the shoot parts of the plants twice with distilled water, dried and ground shoot samples were prepared for elemental analysis using the wet digestion method²⁷.

Statistical analysis

The data were subjected to analysis of variance (ANOVA) using the JMP 13.2 program. It was arranged in a completely randomized design with four replications per treatment. The statistical difference between treatments was determined by the LSD Test at $p < 0.05$. A quadratic regression equation was computed to determine the optimal and toxic B levels for each parameter. The optimum B was estimated by a peak value ($-b/2a$) from the equation ($y = ax^2 + bx + c$), while the toxic dose was estimated as a 50% reduction in the related parameters from the equation: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ ^{28,29}. In the regression formula, the dependent variable, y, was fixed at 50% of the corresponding parameter in the control group.

Results

The analysis of variance showed significant variability in plant height, dry matter, leaf number, leaf area, and seedling fresh and dry weights due to different levels of boron. The results, classified according to their significance level and analyzed by regression analysis, are presented in Fig. 1.

There was a significant reduction in plant height with increasing soil boron levels. The melon plants in control had the tallest plant height, while the shortest height was measured in plants grown in soil with a B

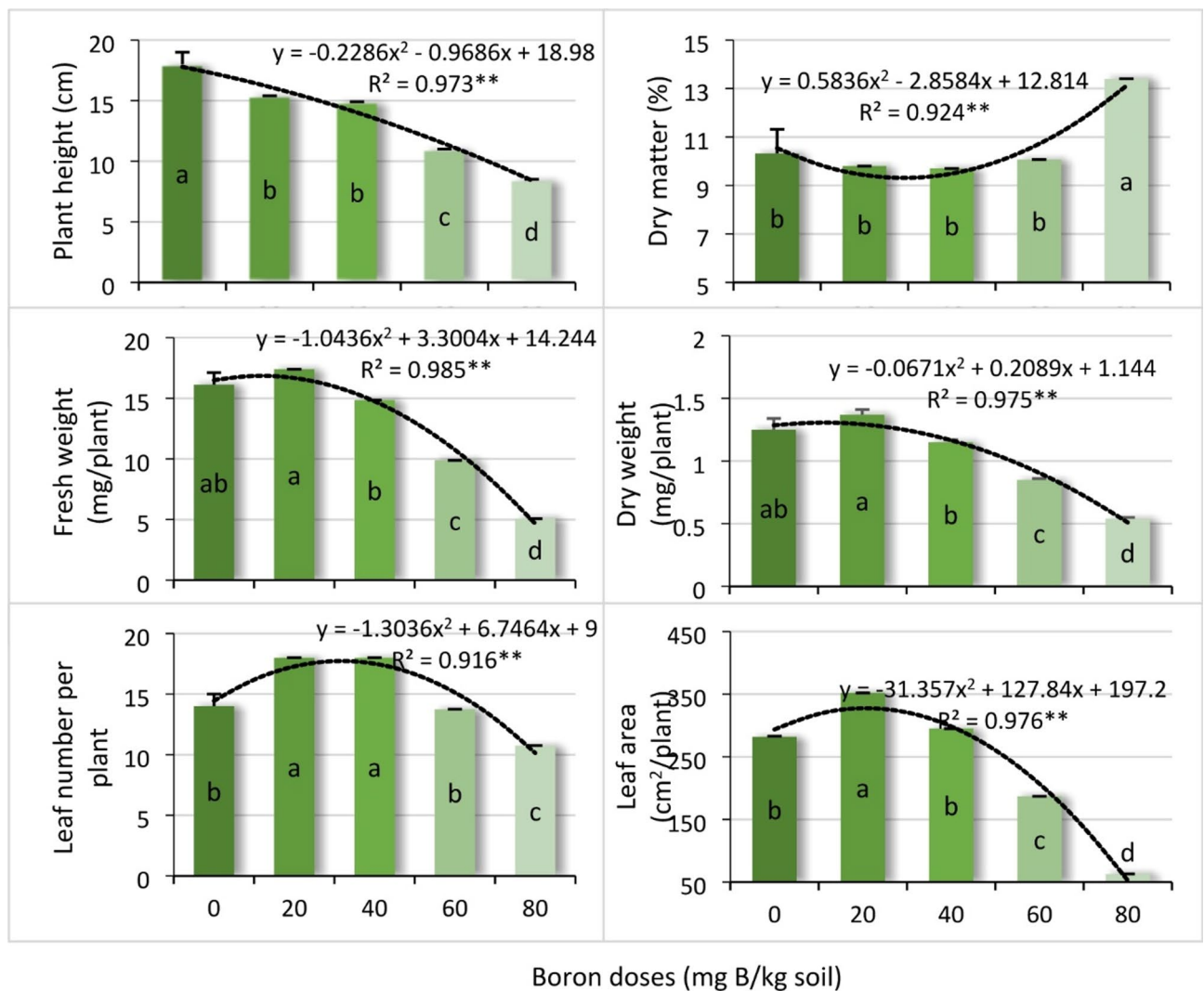


Fig. 1. Changes in plant height, dry matter, fresh weight, dry weight, leaf number and leaf area of melon plants subjected to toxic boron levels. Letter(s) and bar on each column shows the significance level at $p < 0.05$ and standard error, respectively.

concentration of 80 mg/kg. The application of 20 and 40 mg B/kg increased the number of leaves in melon plants, reaching a peak value at 20 mg B/kg. However, it was reduced by the application of 80 mg B/kg. Similarly, leaf area was significantly greater in melon plants grown at a concentration of 20 mg B/kg than in control plants. Concentrations of B higher than 40 mg/kg resulted in a significant decrease in leaf area. The seedling fresh weight was stimulated by 20 mg B/kg, with this concentration yielding the highest value. The dry weight of melon plants increased when they were exposed to a low dose of B (20 mg B/kg), while it decreased when the concentration was increased to 60 mg B/kg. In contrast, no significant changes in dry matter were observed up to 60 mg B/kg, but there was a sharp increase at a concentration of 80 mg B/kg.

Significant differences were found in leaf temperature, chlorophyll index, relative water content, and electrolyte leakage among different levels of B application (Fig. 2). Leaf temperature gradually increased with higher B concentrations, starting at 24.6°C with 0 mg B/kg and peaking at 25.5°C with 60 mg B/kg. However, at 80 mg B/kg, there was a slight decrease in leaf temperature. The chlorophyll index was not significantly influenced by varying levels of B concentrations, ranging from the control to 40 mg/kg. However, it was significantly reduced at 60 mg B/kg and reached the minimum level at 80 mg B/kg. RWC remained relatively stable between 78.3% and 79.5% up to 60 mg B/kg, but dropped to 71.6% at 80 mg B/kg. A concentration of 20 mg B/kg decreased electrolyte leakage from 33.2% at 0 mg B/kg to 27.8% ($p < 0.05$); however, it consistently enhanced with each subsequent increase in B levels. Electrolyte leakage peaked at a concentration of 80 mg of boron per kilogram of soil, indicating a decrease in cell membrane stability.

Figure 3 shows that boron toxicity symptoms occurred at a dose of 40 mg B/kg. The B content in the shoot part of melon plants increased from 65.5 mg B/kg in the control group to 1527 mg B/kg at 80 mg B/kg soil ($p < 0.05$). An increased B concentration in the soil resulted in higher B content in the aboveground parts of the melon plants. Iron (Fe) content declined up to 60 mg B/kg, but surprisingly had the highest value at a B

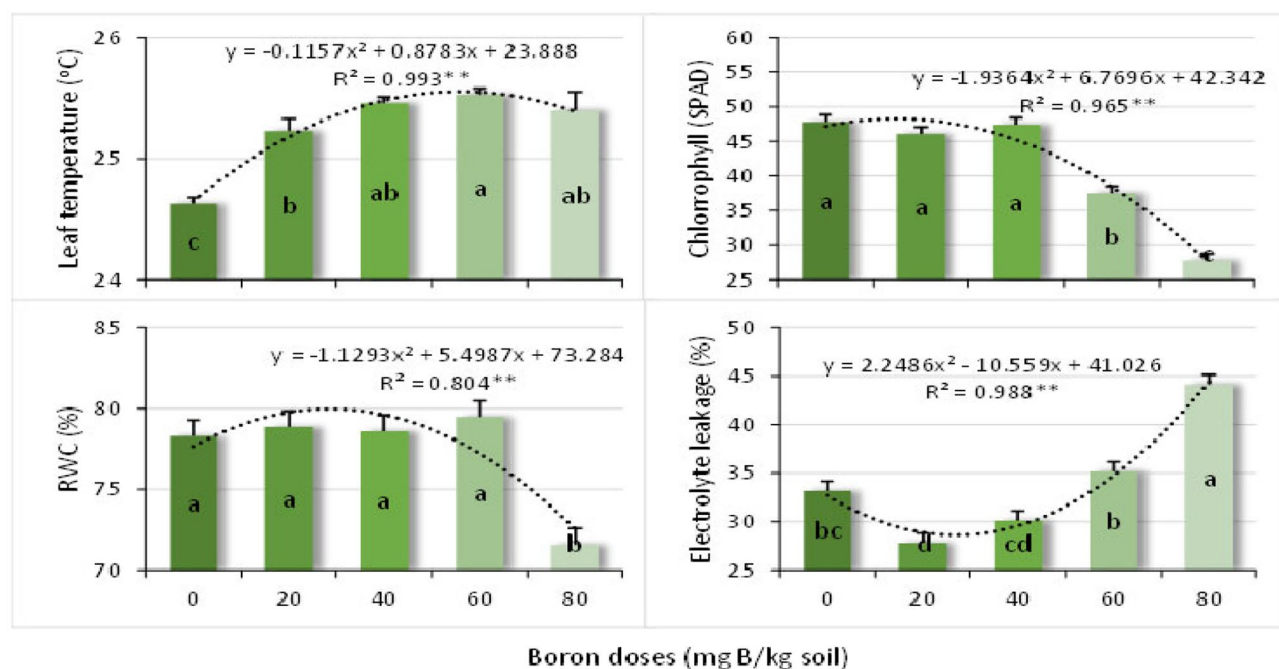


Fig. 2. Changes in leaf temperature, chlorophyll index, relative water content (RWC) and electrolyte leakage of melon plants subjected to toxic boron doses. Letter(s) and bar on each column shows the significance level at $p < 0.05$ and standard error, respectively.

concentration of 80 mg/kg in the soil. Significant changes in manganese (Mn) were observed, with a significant increase at 60 mg B/kg, reaching a peak value of 95 mg Mn/kg at 80 mg B/kg ($p < 0.05$). Similar trends were observed for zinc (Zn) and copper (Cu) contents in melon plants, with the maximum values of Zn and Cu recorded at 80 mg B/kg. Conversely, the residual B content in the soil after crop harvest was strongly related to the applied B levels. The B contents in soils treated with 20, 40, 60, and 80 mg B/kg were determined to be 21.2, 28.2, 36.9, and 50.2 mg B/kg, respectively (Fig. 3).

The relationship between the investigated characteristics was determined by correlation analysis (Table 1). Electrolyte leakage showed significant and negative correlations with leaf area ($r = -0.926^{**}$), seedling fresh weight ($r = -0.906^{**}$), and seedling dry weight ($r = -0.903^{**}$), while it was positively correlated with dry matter ($r = 0.815^{**}$). However, relative water content had the highest correlation with dry matter ($r = -0.905^{**}$). As expected, positive and significant correlations between seedling fresh weight, dry weight, and leaf area were also determined.

*, ** significant at $p < 0.05$ and $p < 0.01$, respectively. PH plant height, LN leaf number, LA leaf area, SFW seedling fresh weight, SDW seedling dry weight, DM dry matter, LT leaf temperature, Chl chlorophyll content, RWC relative water content, EL electrolyte leakage.

Discussion

Increasing boron doses resulted in significant changes in the morphological, physiological, and micronutrient contents of melon seedlings. Higher concentrations of B than 40 mg B/kg inhibited the morphological characteristics of the melon plant, except for plant height, while 20 mg B/kg enhanced them. The toxic level of B for plant height was predicted to be 62.6 mg B/kg soil. Plant length was prominently shortened with increasing doses of B, and this reduction was attributed to the loss of seedling fresh and dry weight. This may be due to shortened internode length, causing reduced plant height even at 20 mg B/kg, as illustrated in Fig. 3. These results are consistent with previous studies on melon³⁰ and [31], and cucumber³², which demonstrated that increasing B treatments significantly reduced shoot fresh and dry weights. In addition, leaf number was significantly lower at 80 mg B/kg compared to 20 and 40 mg B/kg. However, leaf area was sensitive to increasing boron concentrations above 20 mg B/kg, with a significant reduction observed at 60 mg B/kg. This aligns with findings³¹ that reduced leaf area is a response to B stress, and prolonged exposure to B stress resulted in the appearance of visual symptoms in plants. The 50% reduction in leaf area was estimated to occur at 56.5 mg B/kg. The fresh and dry weights of seedlings followed a similar trend with increasing B concentration: they increased at 20 mg B/kg but were significantly reduced at higher B concentrations. The predicted toxic levels of B for both fresh and dry weight were 46.8 mg B/kg and 50.6 mg B/kg, respectively. Restricted root growth was also observed in this study, as shown in Fig. 3. The inhibition of root growth due to excess B may result in a reduction in shoot growth in melon plants by limiting the root area available for water and nutrient uptake. This finding is supported by previous research that reported restricted root growth due to B toxicity, leading to reduced water and B uptake³³.

Increasing B concentration resulted in a higher leaf temperature, which then decreased at 80 mg B/kg soil. This result suggests that melon plants grown under B stress experienced an increase in leaf temperature. The

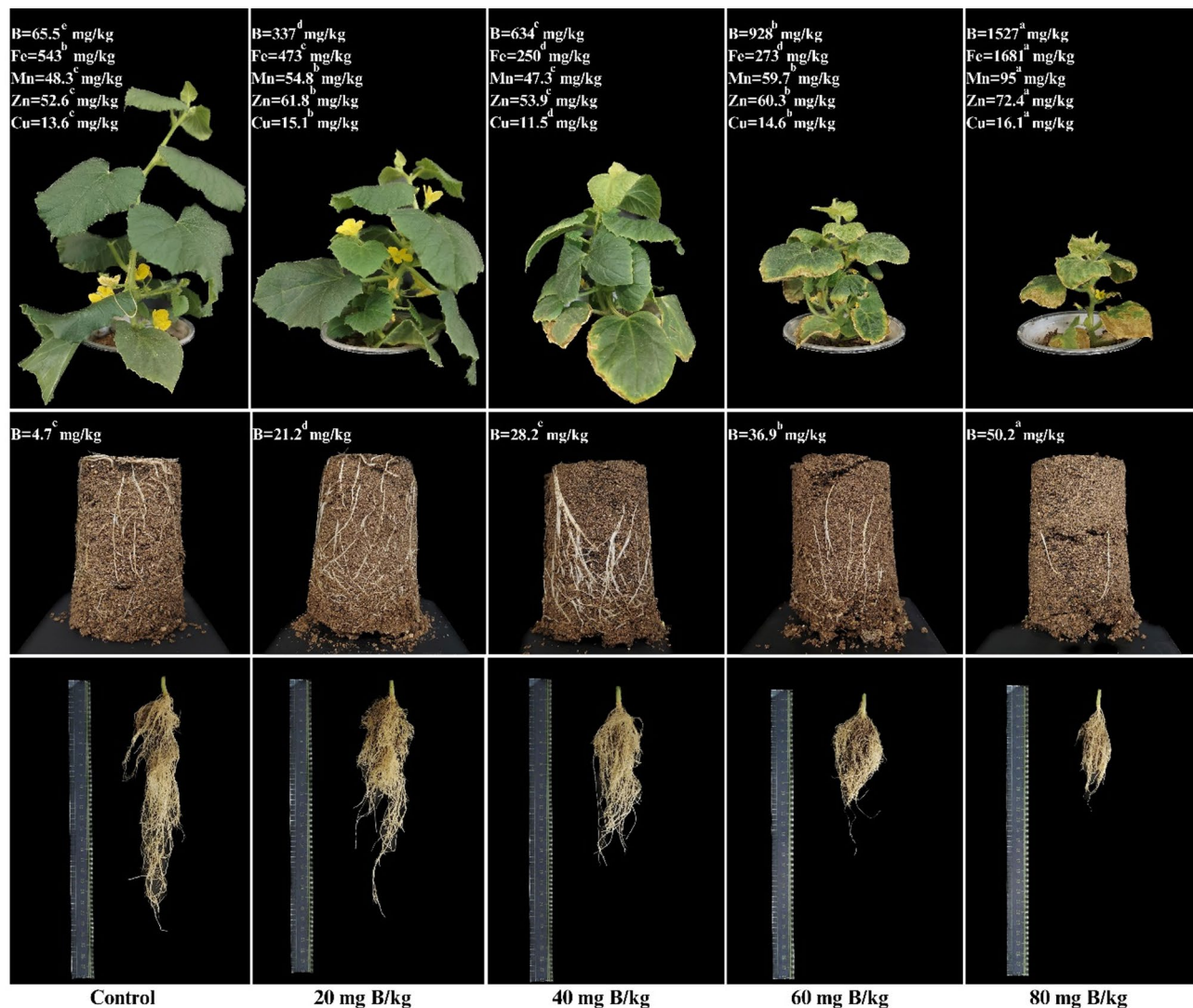


Fig. 3. Changes in some micronutrient contents of aboveground parts (upper row), soil B content (middle row) and root growth (bottom row) of melon plants subjected to toxic boron levels. Letter on each mean shows the significance level at $p < 0.05$.

	LN	LA	SFW	SDW	DM	LT	Chl	RWC	EL
PH	0.563**	0.802**	0.870**	0.853**	-0.596**	-0.601**	0.860**	0.502*	-0.729**
LN	-	0.815**	0.747**	0.735**	-0.752**	0.094	0.730**	0.606**	-0.821**
LA	-	-	0.970**	0.950**	-0.824**	-0.327	0.890**	0.742**	-0.926**
SFW	-	-	-	0.989**	-0.760**	-0.436	0.911**	0.683**	-0.906**
SDW	-	-	-	-	-0.731**	-0.430	0.900**	0.661**	-0.903**
DM	-	-	-	-	-	0.086	-0.756**	-0.905**	0.815**
LT	-	-	-	-	-	-	-0.451*	-0.015	0.195
Chl	-	-	-	-	-	-	-	0.643**	-0.836**
RWC	-	-	-	-	-	-	-	-	-0.703**

Table 1. Correlation coefficients between morpho-physiological characteristics of melon plants exposed to toxic B levels.

chlorophyll index and relative water content responded differently to B levels ($p < 0.01$) because no significant changes were observed up to 40 mg B/kg for Chl and 60 mg B/kg for RWC. A study found a similar trend in apple chlorophyll content, and it was reduced at higher B levels (3 and 6 mM B)³⁴. Similarly, electrolyte leakage was the lowest at 20 mg B/kg soil and significantly higher at 80 mg B/kg soil compared to the control. This finding indicates that 20 mg B/kg acted as a nutrient, reducing electrolyte leakage from leaf tissues due to strong cell membrane stability. Another study conducted by³⁵ showed that the membrane permeability of pepper and tomato did not change significantly between 0 and 5 mg B/kg, but increased considerably at 50 mg B/kg, supporting the present study's finding that 80 mg B/kg destroyed cell membrane stability. Furthermore, increased oxidation and decreased chlorophyll a and b, photosynthetic activity, stomatal conductance, and transpiration rate were observed in *Cucurbita pepo* and *Cucumis sativus* under excess B conditions³⁶. Excess B impaired transpiration rate and caused significant stomata restrictions in loquat³⁷. It is believed that disturbances in these parameters may contribute to growth retardation. Furthermore, the relationships among the investigated characteristics were examined through correlation analysis. It is not surprising that there was a significant relationship between seedling fresh weight, dry weight, and leaf area. However, they were also correlated with electrolyte leakage, which was significantly negatively linked with relative water content, chlorophyll index, leaf number, and plant height. Additionally, a positive correlation between electrolyte leakage and dry matter was observed.

Micronutrient analysis of shoot samples from melon plants grown under different B concentrations showed that increasing B levels negatively affected mineral concentrations. Typically, leaf-burn chlorosis at the margins and tips of older leaves, a symptom of boron toxicity, vanished at 40 mg B/kg (Fig. 3). As expected, shoot B accumulation increased with increasing soil B concentrations. Similarly, soil B concentration was higher with each increase in B. This shows that B absorbed by the root was transported to the shoot as B in the soil increased, as reported by³⁸ in tobacco³⁴, in apple rootstock, and³⁹ in pepper. However, not all boron applied to the soil was absorbed by melon plants, and residual boron remained. For this reason, boron fertilization may affect subsequent plants and overdose application should be avoided. The changes in Fe concentration were significant and its concentration was reduced by increasing B up to 60 mg B/kg soil. A recent study by⁴⁰ showed that B also plays a role in transporting Fe from the roots to the aboveground parts by promoting the reuse of apoplast Fe in *Arabidopsis thaliana* roots. Similar results were reported by³⁸ in tobacco and by³⁴ in apple, where elevated B levels led to increased B concentrations in roots, leaves, and shoots. This suggests that plants absorb more boron when it is present in higher quantities in the soil, even if it reaches a toxic level. Fe concentration was markedly increased at 80 mg B/kg, suggesting a disruption in the ionic balance of the plants. While Mn, Zn, and Cu concentrations were minimally affected by soil B levels up to 60 mg B/kg, they reached their maximum levels at 80 mg B/kg. Mn concentrations fluctuated up to 60 mg B/kg but significantly increased to 95 mg/kg at 80 mg B/kg (Fig. 3). A similar trend was observed in Zn and Cu content, and they were higher at 80 mg B/kg. These findings align with³⁴, who observed that Fe, Mn, and Zn contents in explants decreased as B concentrations in the culture medium increased from 0.1 to 6.0 mM. In the present study, they peaked at 80 mg B/kg, which was an overdose for the survival of the melon plants. According to³⁸, tobacco plants exposed to 20 μ M B showed increased Fe levels and decreased Zn and Cu levels. Post-harvest analysis revealed that higher soil B applications resulted in elevated B concentrations in melon seedlings. Residual B levels were measured at 21.2 mg/kg with a 20 mg B/kg application, and at 50.2 mg/kg with an 80 mg B/kg application (Fig. 3). Higher B levels in the soil correlated with higher B content in the seedlings.

Conclusion

Excess boron in the soil had a negative impact on plant growth, physiological function, and mineral composition in melon plants. The main reason for the inhibition of plant growth seems to be excessive accumulation of B in the shoot, resulting in restricted root, stem, and leaf growth, and disrupted ion balance. This high accumulation of B was closely linked to cell membrane stability and relative water content. On the other hand, the positive effect of 20 mg B/kg could be attributed to its nutritional benefits for melon plant growth. The present study revealed that the harmful effects of B in soil were observed in melon plants at the early growth stage when concentrations exceeded 40 mg B/kg soil.

Data availability

All data generated or analysed during this study are included in this published article.

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

Conflict of interest

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Additional information

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